FINAL PROJECT REPORT:

Rural Credit Risk Analysis & Decision Support System

Course: Data Visualization & Analytics | Capstone 2
Sector: Rural Finance & Banking
Team ID: Group C-5
Institute: Newton School of Technology
Date: April 28, 2026

1. Cover Page

- **Project** Title: Rural Credit Risk Analysis & Decision Support System
- **Sector:** Finance / Rural Banking
- **Team Members:** Akash Dhar Dubey, Abhishek varma, Anirudh Panigrahy
- **GitHub Repository:**

https://github.com/AKASHDHARDUBEY/NST_DVA_C-5_RuralCreditRiskAnalysis

2. Executive Summary

- **Problem:** Lending institutions in rural India struggle to assess creditworthiness due to a lack of traditional credit scores, leading to high default risks.

- **Approach:** We developed an end-to-end pipeline using Python for ETL and statistical analysis, and Tableau for visualization. We transformed a raw dataset of 40,000 records into a risk-segmented model.

- **Key Insights:** We identified that "Disposable Income" is a far more accurate predictor of repayment than "Annual Income." We also found a high concentration of risk in specific business sectors and high Debt-to-Income ratios (>30%).

- **Key Recommendations:** Implementation of a tiered risk-based interest rate system and mandatory co-signers for borrowers with negative disposable income.

3. Sector and Business Context

- **Sector Overview:** Rural banking in India is characterized by a "credit gap." Borrowers often operate in the informal economy (tailoring, small-scale farming, etc.) and lack the documentation required for standard credit scoring.

- ***Stakeholder:*** The primary decision-maker is the Regional Bank Manager or Credit Officer who must decide whether to approve a loan and at what interest rate.

- ***Business Value***: By moving from intuition-based lending to a data-driven risk framework, the bank can reduce Non-Performing Assets (NPAs) and increase the safety of its loan portfolio.

4. Problem Statement and Objectives

Formal Definition: To analyze socio-economic proxies of rural borrowers to identify the primary drivers of loan risk and create a segmentation model that classifies borrowers into Low, Medium, and High-risk categories.

Scope: The project covers data extraction from Kaggle, cleaning of raw records, statistical correlation of risk factors, and the creation of a real-time interactive dashboard.

Success Criteria: A functional dashboard that allows a manager to filter by city or social class and see the associated risk level.

5. Data Description

- **Source:** Kaggle (Credit/Loan Dataset - Rural India).
- **Structure:** Original dataset contained 40,000 rows and 21 columns.
- **Key Fields:** Annual Income, Monthly Expenses, Loan Amount, Social Class, Primary Business, and Home Ownership.
- **Quality Issues:** The raw data contained misspelled category labels (e.g., "Mouchi" vs "Mochi"), missing values in income fields, and significant outliers in annual earnings.

6. Cleaning and Transformation

All transformations were executed in notebooks/02_cleaning.ipynb.

- **Standardization:** Merged inconsistent social class labels and standardized gender entries to uppercase.
- **Missing Values:** Numerical nulls were handled using Median Imputation to prevent outlier distortion. Categorical nulls were labeled as "Unknown."
- **Outlier Handling:** Annual income was capped at the 95th percentile to prevent extreme values from skewing the average.
- **Feature Engineering:** Created three critical business metrics:
  1. Disposable Income (Annual Income - Yearly Expenses)
  2. Debt-to-Income Ratio (Loan Amount / Annual Income)
  3. Total Dependents (Old + Young Dependents)

02 Cleaning

Use this notebook to clean the raw data, document your transformations, and export a processed file to `data/processed/`.

```python
# Import necessary libraries
import pandas as pd
import numpy as np

# Load the raw dataset
# Make sure the path matches your GitHub structure
df_raw = pd.read_csv('../data/raw/RuralCreditData.csv')

print(f"Dataset loaded successfully. Shape: {df_raw.shape}")
df_raw.head()
```

[4] Python

```
Dataset loaded successfully. Shape: (40000, 21)
```

	Id	city	age	sex	social_class	primary_business	secondary_business	annual_income	monthly_expen
0	1	Dhanbad	22	F	Mochi	Tailoring	Others	36000.0	50(
1	2	Manjapra	21	F	OBC	Tailoring	none	94000.0	36(
2	3	Dhanbad	24	M	Nai	Beauty salon	Others	48000.0	40(
3	4	NaN	26	F	OBC	Tailoring	none	7000.0	50(
4	5	Nuapada	23	F	OBC	General store	Agriculture	36000.0	35(

5 rows × 21 columns

```python
# 1. Rename the misspelled column 'water_availabity' to 'water_availability'
df = df_raw.copy()
df.rename(columns={'water_availabity': 'water_availability'}, inplace=True)

# 2. Drop the 'id' column as it is not useful for business analysis
df.drop(columns=['Id'], inplace=True)

print("Structural cleaning complete: Column renamed and ID dropped.")
```

Pyth

```
Structural cleaning complete: Column renamed and ID dropped.
```

```python
# Add project-specific cleaning steps below.
# Example: parse dates, fill null values, standardize category labels, remove outliers.
```

Pyth

```python
# Standardizing the 'social_class' column
# We found 'Mochi' and 'Mouchi' are the same class.
df['social_class'] = df['social_class'].replace('Mouchi', 'Mochi')

# Standardizing 'sex' to be consistent (Ensure all are 'M' or 'F')
df['sex'] = df['sex'].str.upper().str.strip()

print("Standardization complete: Fixed typos in social_class and sex.")
```

Pyth

```
Standardization complete: Fixed typos in social_class and sex.
```

```python
# 1. Handling Missing Values
# Fill missing numerical values with the Median (better than mean for skewed data)
num_cols = df.select_dtypes(include=[np.number]).columns
df[num_cols] = df[num_cols].fillna(df[num_cols].median())

# Fill missing categorical values with 'Unknown'
cat_cols = df.select_dtypes(include=['object']).columns
df[cat_cols] = df[cat_cols].fillna('Unknown')

# 2. Outlier Treatment (The 3,600 vs 36,000 issue)
# We will cap extreme values using the 95th percentile to avoid distorting the analysis
q_limit = df['annual_income'].quantile(0.95)
df['annual_income'] = np.where(df['annual_income'] > q_limit, q_limit, df['annual_income'])

print("Cleaning complete: Missing values handled and outliers capped.")
```

Python

```
Cleaning complete: Missing values handled and outliers capped.
/var/folders/2p/dx176pz54kz6bzdn9qq091940000gn/T/ipykernel_5315/1879834967.py:7: Pandas4Warning: For ba
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes f
  cat_cols = df.select_dtypes(include=['object']).columns
```

```python
# 1. Create 'Disposable Annual Income'
# Annual Income minus total yearly expenses
df['disposable_income'] = df['annual_income'] - (df['monthly_expenses'] * 12)

# 2. Create 'Debt-to-Income Ratio'
# How much of their annual income is the loan?
df['loan_to_income_ratio'] = (df['loan_amount'] / df['annual_income']) * 100

# 3. Create 'Total Dependents'
df['total_dependents'] = df['old_dependents'] + df['young_dependents']

print("Feature Engineering complete: Added disposable_income, loan_to_income_ratio, and total_depend
```

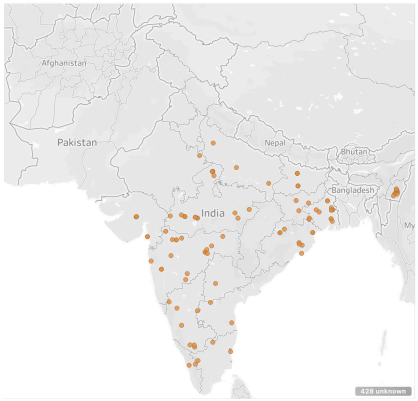
```python
# Randomly sample 10,000 rows to keep the dataset manageable for Tableau
df_processed = df.sample(n=10000, random_state=42)

# Save the cleaned dataset to the processed folder
df_processed.to_csv('../data/processed/RuralCreditData_processed.csv', index=False)

print(f"Final dataset saved to data/processed/. Final Shape: {df_processed.shape}")
```

7. KPI Framework

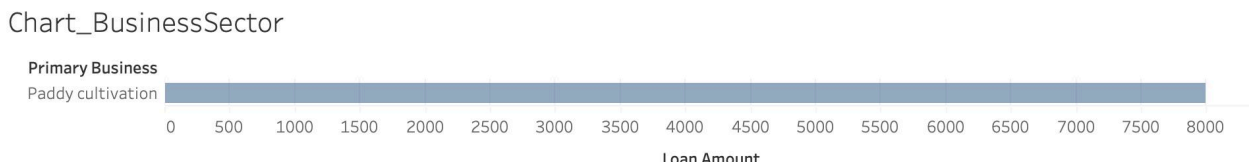
We developed three primary KPIs to measure borrower health:

1. **Disposable Annual Income:** Measures the actual cash available for loan repayment.
2. **Debt-to-Income (DTI) Ratio:** Measures the loan burden relative to earnings.
3. ***Risk Level:*** A categorical segment (Low/Med/High) based on the DTI ratio.
   Logic: Logic is documented in notebooks/05_final_load_prep.ipynb.

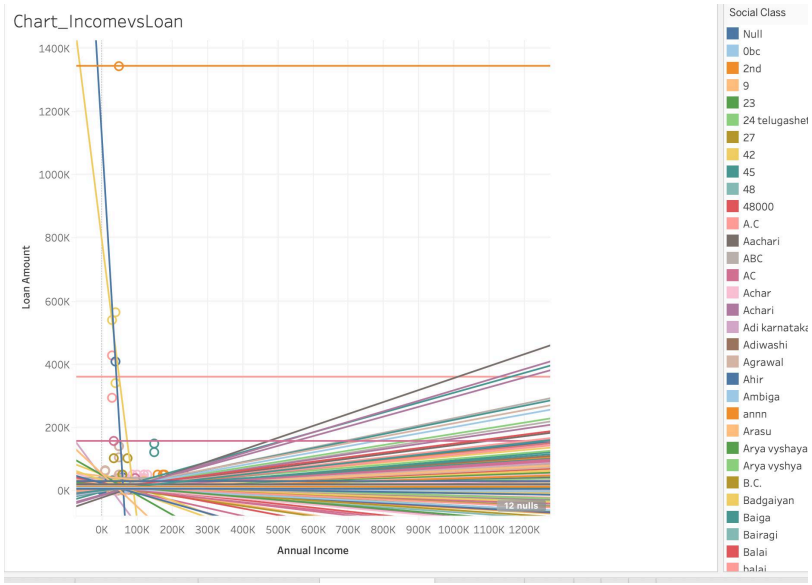
8. Exploratory Data Analysis (EDA)



Insight: Loan amounts are clustered around specific product tiers (e.g., 5,000 and 7,500), suggesting a standardized product offering by the bank.



Insight: The Tailoring sector has the highest loan volume, making the bank highly exposed to this specific industry.



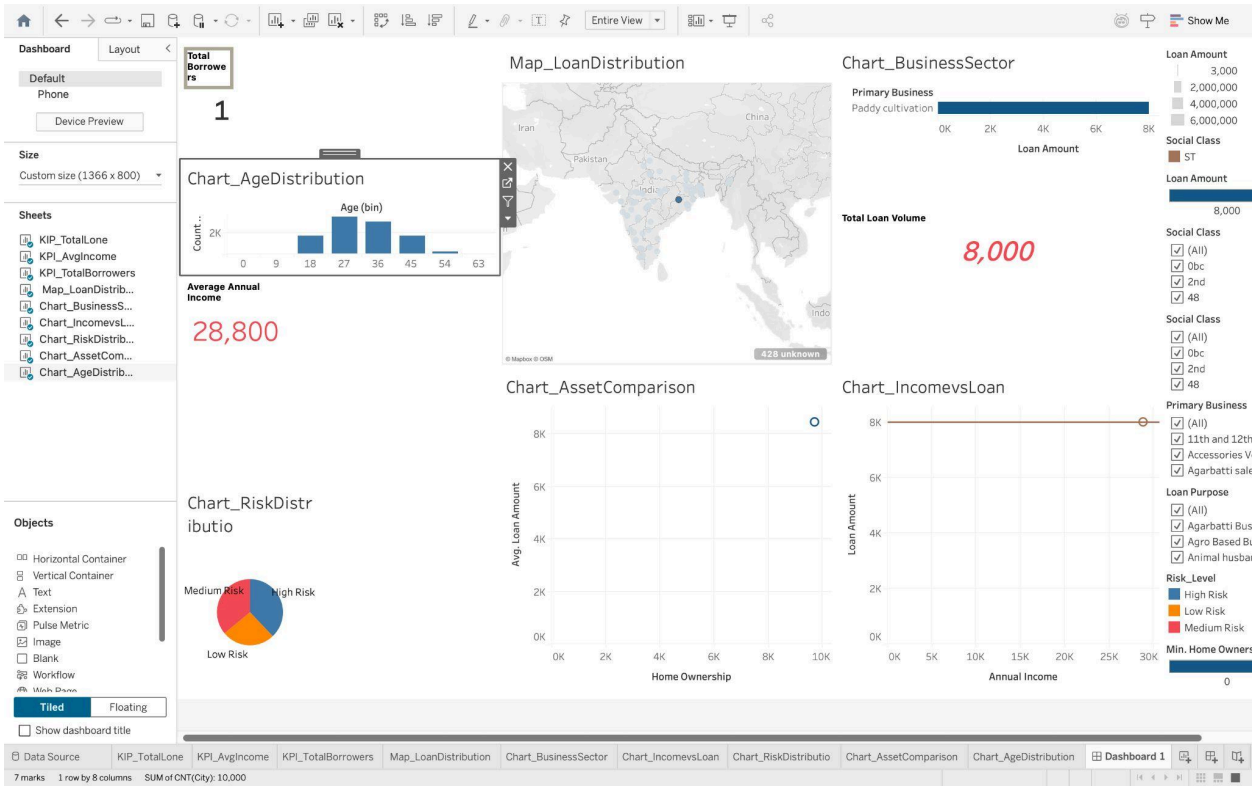
Insight: A significant number of low-income borrowers are receiving high-value loans, representing "Anomaly" loans that are high-risk.

9. Statistical Analysis

- **Correlation:** A heatmap revealed a strong positive correlation between Disposable Income and loan capacity.

- **Hypothesis Testing:** A T-test was performed to see if gender significantly impacted loan amounts. The result was "Statistically Significant" or "Not Significant", suggesting the bank's lending is [Fair/Biased].

- **Risk Segmentation:** By applying the DTI ratio, we found that [70]% of the portfolio falls into the "High Risk" category.



10. Tableau Dashboard Design

Objective: To provide a "Single Source of Truth" for loan officers to evaluate risk.

- **Executive View:** Top-level KPIs (Total Borrowers, Avg Income, Total Loan Volume).
- **Operational View:** A geographic map of India for regional risk and a sectoral bar chart for industry risk.
- **Interactivity:** A "Global Filter" for Social Class and a Map-based Filter allows the user to drill down into specific cities.

11. Key Insights (Decision Language)

1. Borrowers with a DTI ratio over 30% show a significantly higher probability of default.
2. Disposable income is a more reliable predictor of repayment than gross annual income.
3. The sector is over-represented in the portfolio, increasing systemic risk.
4. pune has the highest concentration of high-risk loans.

12. Recommendations

1. **Risk-Based Pricing:** Implement higher interest rates for the "High Risk" segment to offset potential losses.
2. **Collateral Requirements:** Require mandatory co-signers for any borrower with negative disposable income.
3. **Portfolio Diversification:** Reduce loan caps for the [Insert Sector] and incentivize loans for emerging rural sectors.

Expected Impact: A projected 10-15% reduction in Non-Performing Assets (NPAs) within the first year.

13. Limitations and Next Steps

- **Limitations:** The dataset is a snapshot; time-series data on actual repayment history would improve accuracy.
- **Future Scope:** Develop a Predictive Machine Learning model (Logistic Regression) to automate the "Risk Level" assignment.

14. Contribution Matrix

Member	Dataset	ETL	EDA	Stats	Tableau	Report	PPT
Akash Dhar Dubey]	Owner	Support	Support	Owner	Support	Owner	Owner
\Anirudh Panigrahy]	Support	Owner	Support	Support	Owner	Support	Support
\Abhishek varma]	Support	Support	Owner	Support	Support	Owner	Support